Beyond Mapping: A Theory-Grounded Framework for Evaluating Analogical Transfer in Language Models

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Abstract

An accuracy gain after adding an analogy does not establish that a language model transferred its relational structure. Additional explanation, source-answer cues, and changes in prompting can also affect performance, while aggregate gains can conceal damage to previously correct answers. We introduce a theory-grounded framework that makes analogical reasoning observable through four stages: retrieval, mapping, schema induction, and adaptation. Each stage has an explicit information boundary and an auditable output; replacing deficient outputs with reference-informed corrections tests downstream continuation. We support this framework with three studies. Controlled scientific question answering separates analogy gains from explanation and random-example effects. A factorial experiment on narrative reasoning finds that source-answer benefits depend on structural correspondence in several models, with transfer interactions of 14.24 and 12.94 percentage points for Llama-3.1-8B and 70B. Finally, stage interventions reveal that initial retrieval failures can conceal later adaptation failures. The framework distinguishes final-answer correctness, textual reference to an analogy, and successful structural transfer, providing a basis for diagnosing both helpful and harmful analogy use.

1 Introduction

Analogy supports reasoning by carrying a useful relational structure from a familiar source to a new target. In the classic fortress–radiation example, dividing an army across several routes suggests directing multiple low-intensity rays toward a tumor: individual paths remain safe, while their combined effect is sufficient at the target (Gick and Holyoak, 1980). The transferable content is a system of constraints and relations, including convergence and coordination. Sharing a topic or naming corresponding objects is insufficient.

Language models can generate examples, recognize analogies, and sometimes solve problems more accurately after receiving them. Analogical prompting uses self-generated problems and solutions (Yasunaga et al., 2024); teacher-generated analogies can also help scientific concept understanding (Yuan et al., 2024). However, random examples can produce comparable gains on some tasks (Qin et al., 2025). These findings raise a measurement question: *relative to which controls, and through which operations, should an improvement count as evidence of analogical transfer?*

The question has two parts. First, the effect must be attributed carefully. Adding an analogy also adds text, explanation, and potentially answer-relevant knowledge. Comparing only against a prompt without examples cannot separate these factors. Even a positive average gain may combine repairs of incorrect answers with regressions on previously correct ones. Second, the process must be localized. A wrong answer can follow from inadequate sources, inconsistent correspondences, an unusable abstraction, or failure to apply a valid abstraction to target constraints. Final accuracy merges these failures. Conversely, a correct answer may be reached without using the analogy at all.

We address both parts through a theory-grounded framework for evaluating analogical reasoning in language models. Drawing on cognitive accounts of structural alignment, schema induction, and transfer, we operationalize four components: **retrieval**, **mapping**, **schema induction**, and **adaptation**. The model produces a separate output for each stage under an explicit information contract. We diagnose the first inadequate output in an unsuccessful trajectory, replace it with a reference-informed correction, and rerun the downstream stages with the same base model. This distinguishes an initial failure from additional failures that become visible after upstream repair.

The framework is supported by three connected

studies. Phase I examines the attribution of analogy effects in scientific question answering, including helpful and harmful answer transitions. Phase II asks whether a solved, structurally corresponding source provides a selective advantage over a random source. Phase III applies the stage framework to scientific and narrative problems to identify barriers to transfer. These phases answer different questions; together, they link controlled outcome measurement to process diagnosis.

Our contributions are:

- A theory-grounded operational framework with explicit stage outputs and information boundaries, enabling diagnosis and cumulative intervention across retrieval, mapping, schema induction, and adaptation.
- A controlled analysis of analogy effects that distinguishes explanation, example length, relevance, textual mention, and item-level improvements and regressions.
- Empirical evidence for structure-dependent source-answer benefits, followed by a diagnosis showing that repairing early failures can reveal persistent downstream adaptation difficulties.

The stages are an externally imposed experimental decomposition. We neither claim to observe the model's hidden computation nor propose that every analogy must follow this temporal order. Our contribution is to make established theoretical distinctions testable in an auditable language-model workflow.

2 Theoretical Foundations and Related Work

2.1 Analogy involves more than mapping

Structure-mapping theory distinguishes relational correspondence from isolated attribute overlap (Gentner, 1983). A useful mapping preserves role consistency and connected relational systems. Systematicity favors interconnected, particularly causal, relations over disconnected local matches (Clement and Gentner, 1991). Computational structure mapping formalizes constraints on such alignments (Falkenhainer et al., 1989), while alignable differences clarify which discrepancies matter for transfer (Gentner and Markman, 1997).

Access to a source and sound inference from that source are distinct. Surface resemblance can support reminding even when the retrieved source lacks the needed relational structure (Gentner et al., 1993; Holyoak and Koh, 1987). Thus, selecting or generating a plausible example does not establish that it can solve the target. Likewise, a mapping between examples may remain tied to their particular objects. Comparing solved analogues can support a schema that preserves the common solution principle and its conditions of application (Gick and Holyoak, 1983; Gentner et al., 2003). Adaptation then instantiates that schema under the target's own constraints; transfer may fail even when a relevant source is available (Novick and Holyoak, 1991).

These distinctions motivate our four components. They are a synthesis for evaluation, rather than a new claim that cognitive theory contains exactly four universal steps. The framework's methodological contribution is to translate them into explicit artifacts, information boundaries, and interventions. A fluent list of similarities, a general statement of a principle, and a successful target solution are assessed separately.

2.2 Recognizing, generating, and using analogies

Narrative benchmarks such as ARN and AnaloBench assess structural analogy identification under variations in surface similarity and context (Sourati et al., 2024; Ye et al., 2024). Such tasks help establish whether a model can identify a corresponding source, but do not directly show that it can use the source solution to solve a new target. A recent synthesis connects cognitive theories of analogy to NLP research and emphasizes relational understanding beyond entity similarity (Petersen et al., 2026).

For problem solving, analogical prompting generates relevant exemplars before solving the target (Yasunaga et al., 2024). SCUA evaluates the value of teacher-generated analogies for scientific concept understanding (Yuan et al., 2024). These approaches establish practical benefits under their protocols, but source production, source correctness, additional explanation, and target solving can vary together. Qin et al. (2025) directly challenge an interpretation based on relevance alone by showing that random self-generated examples can also help.

Our work builds on these findings in two ways. We first compare analogy conditions with controls targeting specific alternative explanations, rather than treating an accuracy gain as sufficient evidence

of transfer. We then independently vary structural correspondence and source-answer availability. Finally, the stage framework examines the observable intermediates and tests continuation after correcting them. It complements recognition benchmarks and prompting methods by asking where a transfer attempt becomes inadequate and what remains difficult after that failure is repaired.

3 Research Method

3.1 Study design and common evaluation

We organize the research into effect attribution, controlled transfer, and stage diagnosis. The basic comparison unit is a target question with a fixed identifier, choices, and reference answer. Accuracy counts an unextractable answer or a recorded failed response as incorrect. Paired analyses align question identifiers rather than file order. Repeated calls to the same question characterize execution variability; they are not additional independent questions.

Datasets. GPQA Main contains 448 graduate-level scientific questions (Rein et al., 2024): the local version has 78 biology, 183 chemistry, and 187 physics items. We use Main, not the 198-item Diamond subset. MuSR supplies narrative reasoning tasks (Sprague et al., 2024): murder mysteries require combining clues, object placements require tracking locations and agents' knowledge, and team allocation requires satisfying assignment constraints. Unlike GPQA, much of the information needed for MuSR is provided in the narrative. This makes it useful for constructing sources with altered entities but preserved solution relations.

3.2 Phase I: attributing helpful and harmful effects

We compare concept-focused teacher analogies with non-analogical explanations, random analogies, and solver-generated reasoning. The teacher content is generated by DeepSeek V4 Flash. The solvers are Llama-3.1-8B-Instruct, Qwen3-8B, Mistral-7B-Instruct-v0.3, and Gemma-4-31B-it. We compare one 300-word analogy with one 600-word analogy, and compare one, two, and three analogies under a nominal total budget of 600 words. Non-analogical explanations have nominal budgets of 300 or 600 words. Random sources are drawn from the same or a different scientific domain. Chain-of-thought (CoT) and self-analogy prompts provide baselines without an external teacher example (Wei et al., 2022; Kojima et al., 2022).

The controls answer different questions. Length comparisons assess whether more text consistently helps. Explanations assess whether a benefit requires an analogical presentation. Random sources test whether the intended correspondence outperforms unrelated example-shaped content. Multiple examples test the opportunity for comparative abstraction; they do not directly establish that a schema was induced. CoT and self-analogy compare teacher assistance with the solver's own strategies.

For an item-matched baseline b and condition c, we decompose the change in accuracy into

$$p_c - p_b = H_{b\to c} - D_{b\to c}, \tag{1}$$

where H is the fraction of shared items changing from incorrect to correct, and D is the fraction changing from correct to incorrect. Both use the same shared-item denominator. This retains information that a net improvement discards. We also record explicit textual references to the example, analogy, or teacher. A reference can reject the example; neither its presence nor its absence establishes successful structural use.

Llama, Qwen, and Mistral have 448 retained responses per reported condition. Gemma has 435–447, depending on the condition; missing question records are not fabricated, and paired contrasts use the intersection of available identifiers. Its overall accuracies therefore warrant particular care when comparing conditions with different retained sets.

3.3 Phase II: testing structural transfer

For each MuSR target, GPT-5.6-Sol constructs a new source with different entities and context but the intended solution-relevant relational structure. The generator sees the target reference answer to guide source construction; the solver does not receive that answer. Each source contains a narrative, question, choices, and a separate source-answer field. Structural suitability and the absence of an explicit target-answer disclosure are the intended review criteria. Preparation includes format checks, manual review of difficult examples, and a small model-based assessment, rather than exhaustive human validation.

The evaluated set consists of the 103 completed source records available for the main experiment: 61 murder mysteries, 21 object-placement questions, and 21 team-allocation questions. It is not

the full MuSR dataset or a documented representative random sample. Random controls reassign the same sources one-to-one among targets with equal choice counts, with no source assigned to its own target. Assignment reduces same-story and same-task matches where possible and considers length differences. Four five-choice object-placement items retain same-story controls because alternatives are unavailable.

Two factors define four conditions: a corresponding or random source, and a provided or withheld source answer. We denote these by A (corresponding, withheld), B (corresponding, provided), C (random, withheld), and D (random, provided). Withholding removes only the source-answer field; the source narrative, question, and choices remain. The target and instruction are fixed. Sources are labeled neutrally as an "additional example." The maximum output is 2,048 tokens and temperature is zero in all four conditions.

We measure the selective source-answer benefit by

$$\Delta_{\text{transfer}} = (p_B - p_A) - (p_D - p_C). \quad (2)$$

A positive interaction means the source answer is more beneficial when attached to an intended corresponding source. We report all four accuracies: an interaction can arise from a benefit in B, damage in D, or both. A zero interaction does not imply that structural source content has no value.

The four Llama solvers are 3.2-1B, 3.2-3B, 3.1-8B, and 3.1-70B Instruct, each evaluated in three runs. Qwen3-14B and Qwen3-32B provide two-run comparisons outside the Llama family. Run standard deviations describe variability, not confidence intervals. Because random controls permute the same sources, corpus-wide narrative-length distributions are identical (mean 318.99 words), although pairwise and task-specific lengths can differ. We shuffle source choices and update answer indices, then stratify B by whether the source and target correct-answer positions match. This probes index-copying behavior; the strata contain different questions, rather than counterfactual orderings of the same item.

3.4 Phase III: the stage diagnostic framework

The framework uses four fresh model calls. Only the recorded output permitted by each stage contract passes to the next call; conversational state is not carried over. The natural trajectory never receives the target reference answer. Figure 1 summarizes the flow, and Table 1 specifies the contracts.

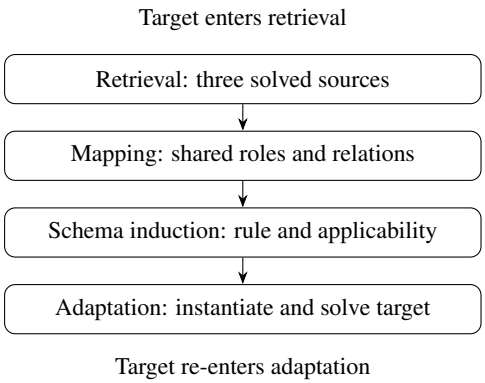


Figure 1: The externally controlled four-stage flow. Mapping receives only retrieval output; schema induction receives only mapping output. Adaptation receives the target and schema.

Operational interpretation. Retrieval names the function of obtaining sources; in this implementation the model can generate them. We do not observe retrieval from internal memory. Mapping aligns the retrieved *sources with each other*. It does not receive the target. Schema induction receives the mapping rather than the raw stories, making explicit whether the intermediate representation is sufficient. Adaptation receives the target again, but not the raw sources or mapping. These restrictions create a traceable source-comparison-to-schema-to-target path. They are experimental choices, not a universal ordering of human analogy.

For example, sources implementing convergence should preserve the constraints that a concentrated force is unsafe along a path, individual weak forces are insufficient, and coordinated forces combine at the goal. Mapping should retain those relations; schema induction should state their conditions without relying on story-specific names; adaptation should check whether the target permits the required combination. "Organize resources better" would be too general to qualify as a useful schema.

Diagnosis and cumulative repair. We inspect naturally unsuccessful trajectories in stage order and record the first output inadequate for its assigned function. A reference-informed author, GPT-6 Sol, constructs a corrected output for that stage while earlier outputs are retained. The author sees the target reference answer, an *open-oracle* construction setting. Upstream corrections must be defensible under the stage's information contract

Stage	Permitted input	Required output and adequacy criterion
Retrieval	Target problem and choices	Three solved source problems, including answers and explanations; sources must support a solution-relevant relational structure.
Mapping	Recorded retrieval output only	Corresponding source roles and relations, source suitability, and material differences; relationships must be consistent and informative.
Schema induction	Recorded mapping output only	An abstract solution rule, roles, relations, applicability conditions, and limits; the schema must preserve enough structure to guide a new solution.
Adaptation	Target and necessary schema content	Instantiate schema roles, check target constraints, make necessary adjustments, and give a final answer supported by the target.

Table 1: Stage contracts operationalize distinctions from cognitive theory. Mapping is across source examples; the schema is connected to the target during adaptation.

and must not explicitly reveal the target answer. The same base solver then reruns only the downstream stages without seeing the reference answer.

If the target is solved, the continuation has been rescued under that correction. Otherwise, we inspect the new trajectory, retain previous corrections, and correct the next inadequate stage. This cumulative procedure reveals failures hidden behind upstream deficiencies. Rescue is conditional on the repaired path and does not estimate the isolated causal effect of each stage across all problems. A gold adaptation output already contains the final answer and has no downstream continuation; its reference agreement is reported separately from base-model rescue.

We report Llama-3.1-8B-Instruct on 100 GPQA Main items and 100 MuSR murder mysteries. Human review covers 30% of questions and associated outputs for logical coherence and correctness. The remaining cases lack human review; oracle-informed construction is not equivalent to exhaustive validation. Diagnosis concerns observed artifacts in the imposed workflow, rather than hidden reasoning mechanisms.

4 Results and Analysis

4.1 Phase I: accuracy gains do not isolate analogy

Table 2 summarizes the reported conditions. A concept-focused 300-word analogy does not outperform both random controls for any of the four solvers. For Llama, a same-domain random source slightly exceeds the intended analogy, while the different-domain control is lower. Qwen matches the same-domain control and trails the different-domain control. Mistral trails both. Gemma's intended analogy lies between the two controls, although its retained sets differ. The results do not establish an analogy-specific advantage merely from receiving a selected example.

Explanation and length. For Mistral, the 300-word teacher analogy improves accuracy over CoT from 16.96% to 27.01%. Yet the non-analogical explanation reaches 27.68%. The benefit therefore does not require an analogical format. In Llama, both teacher inputs trail CoT; in Qwen, self-analogy reaches 43.30%, compared with 42.19% for CoT and 36.61% for the teacher analogy. Doubling analogy length changes accuracy by only 0.45 points for Llama and 0.22 for Qwen. Longer explanations do not uniformly improve performance either. The appropriate baseline depends on the claim: a gain against CoT and a gain against a length-matched explanation answer different questions.

Multiple examples and textual mention. At the same nominal 600-word total budget, additional examples do not produce a monotonic accuracy increase. Mistral reaches 28.79%, 30.58%, and 26.56% with one, two, and three examples. Meanwhile, explicit mentions increase from 12.05% to 19.42% and 29.24%. Multiple examples become more visible in explanations of both correct and incorrect answers. This is consistent with increased engagement with the examples, but does not directly demonstrate successful schema induction.

With the 300-word analogy, Qwen explicitly mentions the example in 31.70% of all responses, versus 8.31% for Gemma, despite Gemma's much higher accuracy. Among Qwen's correct and incorrect responses, mention rates are 31.10% and 32.04%. Textual reference therefore fails to separate successful use from unsuccessful use. Conversely, a correct answer without explicit mention does not establish that the source was ignored.

Helpful and harmful answer changes. Item matching makes the costs of a net gain visible. For Mistral, the 300-word analogy repairs 86 CoT errors among the 448 items (19.20% of all items), but turns 41 previously correct answers into errors

Condition	Llama-8B	Qwen-8B	Mistral-7B	Gemma-31B
Analogy, 300 words	30.80	36.61	27.01	65.39
Analogy, 600 words	31.25	36.83	28.79	68.69
Two analogies, 2×300 words	31.92	35.27	30.58	69.00
Three analogies, 3×200 words	31.03	39.73	26.56	69.61
Non-analogical explanation, 300 words	31.03	37.72	27.68	68.25
Non-analogical explanation, 600 words	30.58	36.61	25.89	68.47
Random same-domain analogy, 300 words	31.25	36.61	27.68	64.85
Random different-domain analogy, 300 words	29.69	39.73	28.13	67.12
Chain-of-thought	32.14	42.19	16.96	66.52
Self-analogy	28.79	43.30	22.54	64.06

Table 2: Phase-I accuracy (%) on GPQA Main. Llama, Qwen, and Mistral have 448 retained responses per condition. Gemma has 435–447, so its overall contrasts can also reflect changes in the retained question set. Recorded responses without a final answer are incorrect.

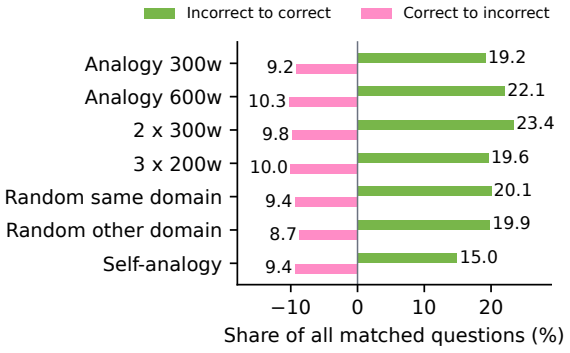


Figure 2: Mistral transitions relative to CoT on the same 448 questions. Improvements and regressions use all shared items as denominator. Their difference is the net accuracy change.

(9.15%). The net gain is 45/448, or 10.04 percentage points. Two 300-word analogies repair 105 answers and damage 44. Figure 2 shows that improvements and regressions occur together across conditions. These are condition-associated transitions; differing instructions and comparable transitions under random controls prevent attributing every regression to a faulty analogical mapping.

These observations motivate sources constructed around the target's solution relations, rather than concept explanations alone. Reference-informed construction in the following phases aims to improve the guide's structural adequacy; it does not give the base solver the target answer.

4.2 Phase II: source answers help selectively

Table 3 reports the factorial cells and their interaction. Llama-1B gains 5.50 points from providing either source answer, giving zero mean interaction. Llama-3B benefits from corresponding source content by about 13.92 points in both answer settings, but providing the answer produces the same 2.91-point gain in corresponding and random sources. Thus, zero interaction can coexist with a substantial source-content advantage.

The two larger Llama models show a different pattern. For 8B, the corresponding source answer increases accuracy by 13.59 points while the random answer changes it by -0.65, giving a 14.24-point interaction. For 70B, the corresponding answer adds 12.94 points and the random answer adds none. These results support selective use of the intended corresponding solution under this protocol. They do not imply that every correct answer used the analogy or that transfer increases monotonically with model size.

Both Qwen models show positive interactions as well. Qwen-14B benefits from the corresponding answer while its random-answer mean is unchanged. For Qwen-32B, the interaction combines a 5.34-point corresponding-answer gain with a 7.28-point random-answer loss. Reporting the cells matters: the larger interaction does not mean a larger benefit from the correct guide.

Answer roles versus answer positions. Llama-1B obtains 88.24% accuracy when correct source and target answer positions match, but only 0.64% when they differ. Together with its identical answer effects for corresponding and random sources, this suggests predominant index following. Llama-8B and 70B retain 57.05% and 65.38% accuracy in the different-position stratum; their same-position accuracies are 74.51% and 79.08%. These results are compatible with more than index copying, although position sensitivity persists. Because the strata contain different target items, we interpret this check alongside the factorial comparison, not as a standalone causal estimate.

Model	A: true/no answer	B: true/answer	C: random/no answer	D: random/answer	Δ_{transfer}
Llama-3.2-1B	38.51 ± 1.48	44.01 ± 0.56	38.51 ± 2.02	44.01 ± 0.56	0.00
Llama-3.2-3B	54.05 ± 1.48	56.96 ± 0.56	40.13 ± 1.12	43.04 ± 1.48	0.00
Llama-3.1-8B	52.10 ± 3.68	65.70 ± 3.41	52.43 ± 1.94	51.78 ± 7.35	+14.24
Llama-3.1-70B	59.22 ± 3.88	72.17 ± 8.76	53.07 ± 2.24	53.07 ± 2.02	+12.94
Qwen3-14B	66.02 ± 4.12	75.24 ± 0.69	60.68 ± 2.06	60.68 ± 3.43	+9.22
Qwen3-32B	68.93 ± 2.75	74.27 ± 0.69	65.53 ± 2.06	58.25 ± 2.75	+12.62

Table 3: Phase-II accuracy (%, mean $\pm$ run SD) on the same 103 MuSR targets in every cell. Llama has three runs; Qwen has two. Interactions are in percentage points, computed from unrounded run-level values; rounding displayed cells can produce small discrepancies. SD is not an inferential confidence interval.

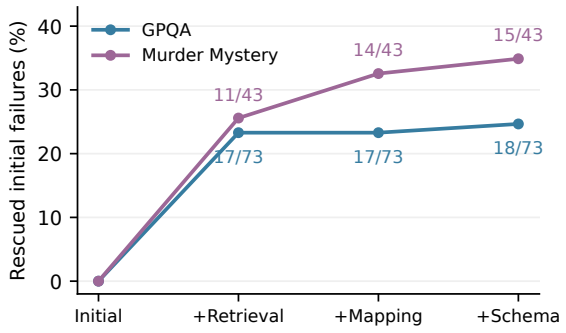


Figure 3: Cumulative rescue of initially failed workflows after upstream corrections. Counts use the actual cohorts: 73 GPQA and 43 murder-mystery failures. Final gold adaptation is excluded because it supplies the answer rather than testing downstream base-model continuation.

4.3 Phase III: locating and revealing failures

We now ask what prevents a transfer attempt from succeeding. The natural four-stage Llama-8B workflow answers 27/100 GPQA items and 57/100 murder mysteries correctly. All 73 unsuccessful GPQA paths are first diagnosed at retrieval. For murder mysteries, the 43 failures begin at retrieval in 19 cases, mapping in 3, and adaptation in 21; none is initially assigned to schema induction. These distributions are conditional on failed workflow outputs, not stage failure rates over all items.

First failure is not the full diagnosis. After reference-informed retrieval correction, the base model solves 17 of the 73 GPQA failures (23.29%). Mapping corrections do not increase this total; cumulative schema corrections add one rescue, bringing it to 18/73 (24.66%). For murder mysteries, cumulative upstream corrections rescue 11, then 14, then 15 of the 43 original failures after retrieval, mapping, and schema interventions (25.58%, 32.56%, and 34.88%). Figure 3 tracks the same initial failure cohorts throughout.

After upstream repair, adaptation remains inadequate for 55/73 GPQA failures (75.34%) and 28/43 murder mysteries (65.12%). Across the cumulative trajectories, GPQA additionally reveals mapping problems in 26/73 and schema problems in 15/73. Murder mysteries reveal mapping problems in 6/43 and schema problems in 1/43. These counts overlap: one trajectory can need several corrections. In particular, diagnosing all initial GPQA failures at retrieval does not mean that improving retrieval eliminates the transfer problem.

Including natural successes and genuine base-model continuations after upstream repair yields 45/100 correct GPQA answers and 72/100 correct murder-mystery answers. These are assisted workflow outcomes, not independent solver accuracies. Replacing the final adaptation with an oracle-informed answer is excluded from both totals.

The domain contrast is informative. Scientific questions require external knowledge both to understand the target and to produce useful sources; narrative mysteries supply many relevant facts in the input but require applying relations under detailed target constraints. The observed diagnoses are compatible with this distinction. The intervention results show why both access to useful sources and reliable target adaptation must be assessed: the first visible obstacle can conceal a later one.

5 Discussion and Conclusion

The three studies distinguish practical benefit from evidence about transfer. Phase I shows that an analogy condition can improve average accuracy without outperforming an explanation control, and can repair some answers while damaging others. The benefit is real under that comparison, but its interpretation depends on what the baseline controls. Textual mention adds a behavioral observation without resolving whether the relevant relation was used correctly.

Phase II provides a more targeted test. The source answer helps several models more when attached to a structurally corresponding source than to a reassigned source from the same corpus. This supports structure-dependent use of source solutions under the tested construction and prompting protocol. It also distinguishes three behaviors: index following, benefits from corresponding source content without a selective answer effect, and selective source-answer benefits. A single accuracy ranking would obscure these differences.

The framework connects these outcome distinctions to inspectable operations. Its central contribution is the combination of theoretical stage distinctions, explicit information contracts, and continuation tests after intermediate corrections. Cognitive theory motivates evaluating source access, relational alignment, abstraction, and target application separately. The workflow makes these distinctions executable without asserting that its generated intermediates expose the model's internal computation.

The repairs reveal a useful diagnostic distinction: *where a path first fails* differs from *which capabilities remain inadequate after repair*. GPQA initially appears to have a retrieval-only problem, yet most of its failed paths remain unsuccessful after useful upstream information is supplied. For murder mysteries, adaptation is prominent both initially and after upstream repair. Thus, better sources alone are insufficient; a model must preserve the useful structure and apply it under the target's constraints.

Future work can compare this information flow with alternative stage interfaces, expand source validation, and test robustness across model families and domains. Matched malformed-stage controls would further separate the value of a specific correction from generic benefits of additional guidance. The present evidence supports an evaluation principle: measure helpful and harmful outcome changes, test whether source-answer benefits depend on correspondence, and inspect the stages needed to turn a source into a target solution. This gives a more precise account of analogical reasoning than accuracy or mapping quality alone.

Limitations

The framework diagnoses an imposed, text-based workflow. Its intermediates are behavioral artifacts, not direct observations of hidden model computation. Requiring fresh calls and restricting information may introduce difficulties absent from unconstrained solving. The four-stage synthesis is one operationalization of cognitive distinctions; alternative interfaces or orderings may yield different diagnoses.

Phase I controls particular alternative explanations but does not isolate every source of condition differences. Instructions differ between teacher-assisted and solver-generated conditions, and nominal word budgets do not guarantee equal token or computational budgets. Gemma's retained question counts vary across conditions. Overall differences for that model can therefore include changes in question composition; matched-item comparisons require shared identifiers. Recorded invalid answers and service failures can also influence accuracy. Llama's recovered Phase-I records combine an initial DeepInfra run with AvalAI recovery of unsuccessful requests.

Phase II uses 103 completed synthetic-source records rather than a representative sample of all MuSR questions. Human source validation is incomplete. Correspondence is an intended and partially reviewed construction property; random reassignment removes the designed correspondence but cannot guarantee the absence of incidental useful relations. The four same-story object-placement controls and task-specific length differences further limit isolation. Global narrative-length matching does not match each pair's information content. The experiment manipulates source-answer *availability*, not correct versus deliberately incorrect source answers.

The model-size comparison also changes Llama generation, provider, and sometimes quantization. The 1B and 3B models use Cloudflare, 8B uses Novita FP8, and 70B uses DeepInfra; Qwen uses DeepInfra FP8. Consequently, these observations do not identify a pure parameter-count effect. Two or three repeated executions describe run variability rather than population uncertainty. Object-placement questions can share stories, and comparisons within answer-position strata do not control item difficulty. We make no significance claim or universal scaling claim from these results.

Phase III covers one reported base model and 100 questions per dataset. Diagnosis focuses on unsuccessful staged outputs and does not estimate unconditional failure rates or spontaneous analogy use. Human review covers 30% of questions and associated outputs, without a reported inter-annotator agreement estimate. GPT-6 Sol authors corrections with access to the reference answer. Even without explicit answer disclosure, answer-informed sources or schemas can carry strong solution guidance. Rescue is therefore an assisted behavioral outcome, conditional on the oracle-informed artifact and the preceding correction history. Final-answer replacements are not independent model successes. These limits motivate reporting natural performance, cumulative upstream rescue, and oracle answer agreement separately.

Ethical Considerations

We study existing scientific and synthetic narrative benchmarks and model-generated sources. Improvements in benchmark performance do not establish reliable scientific or educational advice. An analogy can be persuasive while omitting a decisive constraint; the observed answer regressions reinforce the need to check applicability before deploying analogy-based assistance. Dataset and model access remain subject to their respective licenses and service terms. Generative AI assisted conversion of the research manuscript into English prose and LaTeX; the authors remain responsible for checking the claims, citations, numerical results, and final text. The use of models to generate sources and reference-informed corrections is described in the method.

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A Execution and Evaluation Details

This appendix summarizes the retained experimental protocol. It describes stage tasks and condition construction rather than presenting reconstructed wording as the exact historical prompts.

A.1 Phase-I condition identifiers

The reported 300-word analogy is condition 1, focused on the main scientific concept. Condition 2 uses one 600-word analogy; conditions 3 and 4 use two 300-word and three 200-word analogies. Conditions 5 and 6 are non-analogical explanations of nominal length 300 and 600 words. Conditions 7 and 8 are random same-domain and different-domain analogies. Condition 20 requests CoT and condition 21 requests solver-generated analogy, with a source-generation ceiling of 600 words. The broader study also contains condition 0, an unrestricted-concept analogy; it is not the selected main analogy in Table 2.

Available Gemma response counts for conditions 1, 2, 3, 4, 5, 6, 7, 8, 20, and 21 are respectively 445, 444, 442, 441, 441, 444, 441, 444, 435, and 447. Llama, Qwen, and Mistral have 448 retained responses in each reported condition. Correctness transitions are computed by identifier intersection, with recorded missing final predictions treated as incorrect. A missing question record is distinct from a recorded response lacking a valid answer.

A.2 Phase-II source construction and assignment

Sources contain `analogy`, `question`, `choices`, and `answer` fields. Generation preserves the target choice count, requests a single correct

source option, and caps the source narrative below 1,000 words. Source-choice reordering uses seed 20260901; random-control assignment uses seed 20260903, with the source answer updated after reordering. Withholding the answer removes only `answer`. The older exploratory no-solution condition also withheld source choices and is not used as the main factorial comparison.

Mean corresponding/random narrative lengths by task are 371.41/371.41 words for murder mysteries, 189.14/228.76 for object placements, and 296.57/256.95 for team allocation. Their weighted corpus-wide means are both 318.99 words. Four five-choice object-placement questions have same-story random controls. This is a remaining source-construction limitation rather than evidence of complete structural independence.

A.3 Interpreting the answer-position check

In the source-answer condition B, 51 of the 103 questions have matching source and target answer indices and 52 have differing indices. Pooling three Llama runs yields 153 and 156 response events, respectively; it does not yield that many independent questions. The counts of correct matching/differing events are 135/1 for 1B, 93/83 for 3B, 114/89 for 8B, and 121/102 for 70B. These counts reproduce the position-stratified accuracies in Section 4.2.

B Diagnostic Cohorts and State Accounting

Table 4 reports the actual Phase-III counts. Some retained visualization assets display counts multiplied by two. That display transformation preserves percentages but adds no observations; the paper uses the original 100-item dataset cohorts and the 73/43 initial failure cohorts throughout.

The stage burdens accumulated across interventions overlap, unlike the mutually exclusive states in each row. GPQA has 73 paths needing retrieval correction, 26 needing mapping correction, 15 needing schema correction, and 55 reaching inadequate adaptation. For murder mysteries, these counts are 19, 6, 1, and 28. A path can contribute to several of these totals; summing them does not count failed questions.

The saved Phase-III base-model identifier is `cf.llama-3.1-8b-instruct-fast`. No completed second-model comparison is included in this manuscript. Natural successes are 27 GPQA and 57 murder-mystery questions; cumulative genuine upstream rescue contributes 18 and 15, respectively. The agreement of reference-informed final outputs is distinct from these base-model continuations.

State	Rescued	R	M	S	A
GPQA: 73 initial failures					
Initial	0	73	0	0	0
After R	17	0	26	7	23
After M	17	0	0	15	41
After S	18	0	0	0	55
Murder mysteries: 43 initial failures					
Initial	0	19	3	0	21
After R	11	0	6	0	26
After M	14	0	0	1	28
After S	15	0	0	0	28

Table 4: Actual cohort counts over cumulative repair. Each row partitions the same initial failures into rescued paths and the current first inadequate stage: retrieval (R), mapping (M), schema (S), or adaptation (A). Final gold adaptation is excluded.